Machine Learning

Lecture 5: From the Normal Equation to PCA

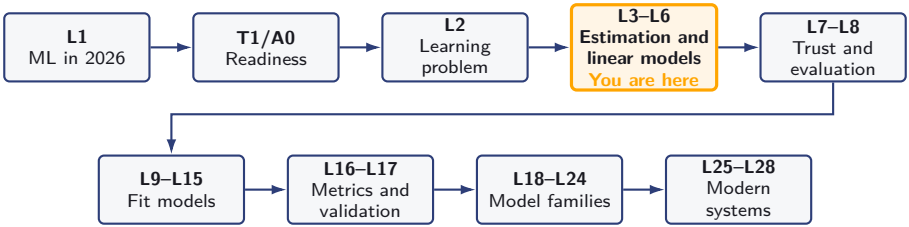
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Civil Engineering; Computer Science and Engineering

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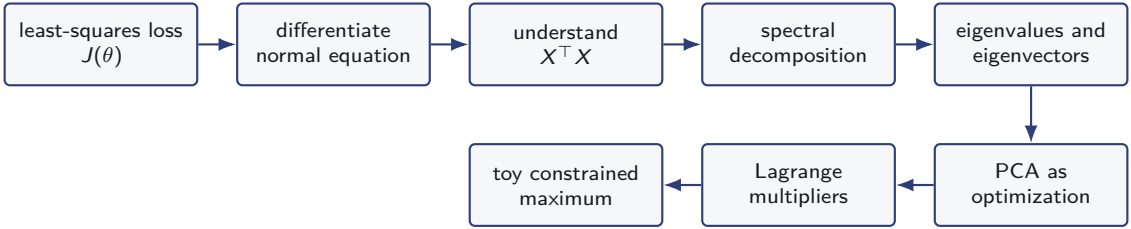


Course Roadmap: You Are Here



Lecture 4B established least squares and projection. Today we reconstruct the normal equation carefully, use spectral decomposition to expose its coefficient directions, and then reuse the same eigenvalue language to formulate PCA as constrained optimization.

What We Actually Built On The Board Today



Summary

The lecture has one continuous question: how does a matrix reveal the directions that the data determine strongly, weakly, or not at all?

Learning Objectives

By the end of this lecture, you should be able to:

- derive $\nabla J(\theta) = X^\top (X\theta - y)$ without guessing the transpose;
- obtain and interpret the normal equation $X^\top X\widehat{\theta} = X^\top y$;
- explain why $X^\top X$ is symmetric and positive semidefinite;
- use the spectral theorem to describe regression direction by direction;
- explain where $a = Q^\top \widehat{\theta}$ and $b = Q^\top X^\top y$ come from;
- formulate PCA by maximizing projected variance;
- derive a Lagrange-multiplier system and reuse it to obtain $Cq = \lambda q$.

Notation: Keep Track Of The Space Before The Algebra

Object	Shape	Meaning
X	$n \times d$	n observations, d predictors
y	$n \times 1$	observed response
θ	$d \times 1$	regression coefficients
$X\theta$	$n \times 1$	fitted response
$r = y - X\theta$	$n \times 1$	residual vector
$X^\top X$	$d \times d$	predictor Gram matrix
q_j	$d \times 1$	one eigenvector direction; matrix specified in context
X_c	$n \times d$	centred feature matrix for PCA
q	$d \times 1$	one candidate PCA feature direction
$z = X_c q$	$n \times 1$	projected coordinate for every observation
$C = \frac{1}{n-1} X_c^\top X_c$	$d \times d$	sample covariance matrix

Summary

Regression moves a coefficient vector θ. PCA chooses a feature direction q and produces scores $z = X_c q$. The direction and the scores live in different spaces.

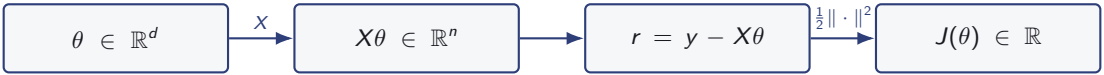
Start From The Least-Squares Objective

For observation i,

$$\widehat{y}_i = x_i^\top \theta, \qquad r_i = y_i - x_i^\top \theta.$$

Add the squared residuals:

$$\boxed{J(\theta) = \frac{1}{2}\sum_{i=1}^{n}(y_i - x_i^\top \theta)^2 = \frac{1}{2}\|y - X\theta\|_2^2.}$$



Expand The Objective Before Differentiating

Use $(y - X\theta)^\top (y - X\theta)$:

$$\begin{aligned} 2J(\theta) &= (y - X\theta)^\top (y - X\theta) \\ &= y^\top y - y^\top X\theta - \theta^\top X^\top y + \theta^\top X^\top X\theta. \end{aligned}$$

The two middle terms are the same scalar:

$$y^\top X\theta = (y^\top X\theta)^\top = \theta^\top X^\top y.$$

Therefore

$$\boxed{J(\theta) = \frac{1}{2} y^\top y - \theta^\top X^\top y + \frac{1}{2} \theta^\top X^\top X\theta.}$$

Summary

The transpose appears because the derivative must return a d-dimensional coefficient-space vector.

Differentiate One Coefficient At A Time

For coefficient θ_j, begin with

$$J(\theta) = \frac{1}{2} \sum_{i=1}^{n} \left(y_i - \sum_{k=1}^{d} x_{ik} \theta_k \right)^2 .$$

Apply the chain rule and identify the jth component of the gradient.

Let $r_i(\theta) = y_i - \sum_{k=1}^{d} x_{ik} \theta_k$. Then

$$\frac{\partial r_i}{\partial \theta_j} = -x_{ij}.$$

Hence

$$\begin{aligned} \frac{\partial J}{\partial \theta_j} &= \frac{1}{2} \sum_{i=1}^{n} 2 r_i \frac{\partial r_i}{\partial \theta_j} \\ &= -\sum_{i=1}^{n} x_{ij} r_i \\ &= \sum_{i=1}^{n} x_{ij} (x_i^\top \theta - y_i). \end{aligned}$$

This is component j of $X^\top (X\theta - y)$.

Stack The Derivatives: The Normal Equation Appears

Stack the d scalar derivatives, set the gradient equal to zero, and rearrange. Check the dimensions of both sides.

Stacking all components gives

$$\boxed{\nabla J(\theta) = X^\top (X\theta - y)} \in \mathbb{R}^d.$$

At a stationary point $\widehat{\theta}$,

$$X^\top (X\widehat{\theta} - y) = 0$$
$$X^\top X\widehat{\theta} - X^\top y = 0$$
$$\boxed{X^\top X\widehat{\theta} = X^\top y}.$$

Both sides are vectors in $\mathbb{R}^d$.

Normal equation: one scalar stationarity equation for each coefficient.

The Same Normal Equation Is An Orthogonality Statement

Define the fitted residual

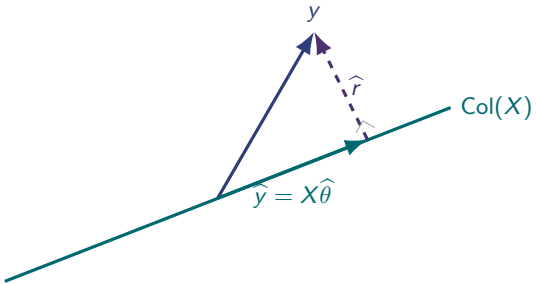
$$\widehat{r} = y - X\widehat{\theta}.$$

The normal equation can be written as

$$\boxed{X^\top \widehat{r} = 0.}$$

If x_j is column j of X, then component j says

$$x_j^\top \widehat{r} = 0.$$



When May We Write The Inverse Formula?

From

$$X^\top X\widehat{\theta} = X^\top y,$$

we may write

$$\boxed{\widehat{\theta} = (X^\top X)^{-1} X^\top y}$$

only when $X^\top X$ is invertible.

Full column rank

$\operatorname{rank}(X) = d$. No nonzero coefficient move v satisfies $Xv = 0$. The least-squares coefficients are unique.

Warning

If predictors are exact linear combinations, $X^\top X$ is singular. The normal equation still holds, but the inverse formula does not.

What Information Is Stored In $X^\top X$?

Write the design matrix by columns:

$$X = [x_1 \ \cdots \ x_d].$$

Then

$$\boxed{X^\top X = \begin{bmatrix} x_1^\top x_1 & \cdots & x_1^\top x_d \\ \vdots & \ddots & \vdots \\ x_d^\top x_1 & \cdots & x_d^\top x_d \end{bmatrix}.}$$

Diagonal entry $x_j^\top x_j = \|x_j\|^2$: the squared size of predictor j.

Off-diagonal entry $x_j^\top x_k$: the overlap between two predictor patterns.

Summary

$X^\top X$ is the Gram matrix of the predictor columns. Correlated columns appear as strong off-diagonal coupling.

Why $X^\top X$ Is Positive Semidefinite

Take any coefficient direction $v \in \mathbb{R}^d$:

$$\begin{aligned} v^\top (X^\top X) v &= (Xv)^\top (Xv) \\ &= \|Xv\|_2^2 \\ &\geq 0. \end{aligned}$$

Therefore

$$\boxed{X^\top X \succeq 0.}$$

For least squares,

$$\nabla^2 J(\theta) = X^\top X,$$

so the objective is convex in θ.

Summary

Equality holds exactly when $Xv = 0$: moving coefficients along v changes no fitted value. Strict positivity for every nonzero v gives a unique minimizer.

Why Eigenvectors Enter Regression At This Point

Suppose rainfall and wetness predictors are strongly coupled:

$$X^\top X = \begin{bmatrix} 10 & 9.8 \\ 9.8 & 10 \end{bmatrix}.$$

The original coefficient axes mix the two effects. Ask instead:

Question

Are there special coefficient combinations that $X^\top X$ acts on without turning them into a different direction?

Those directions satisfy

$$\boxed{(X^\top X)q = \lambda q.}$$

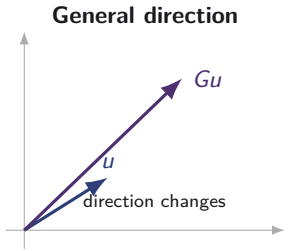
q: a coefficient combination, such as increasing both coefficients together.

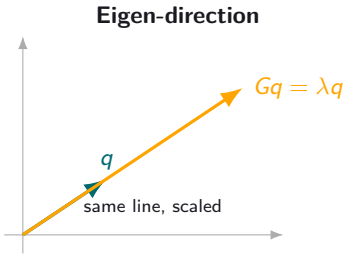
λ: how strongly the data and loss react to that combination.

Eigenvector Means Same Line After The Matrix Acts

A nonzero vector q is an eigenvector of a square matrix G when

$$\boxed{Gq = \lambda q.}$$





Worked Eigenpair Example

For

$$G = \begin{bmatrix} 5 & 3 \\ 3 & 5 \end{bmatrix},$$

find the eigenvalues and one unit eigenvector for each. Identify the strong and weak directions.

$$\det(G - \lambda I) = \det \begin{bmatrix} 5-\lambda & 3 \\ 3 & 5-\lambda \end{bmatrix}$$
$$= (5-\lambda)^2 - 9 = 0.$$

Thus $\lambda_1 = 8$ and $\lambda_2 = 2$.

$$q_1 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ 1 \end{bmatrix}, \qquad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

Check: $Gq_1 = 8q_1$ and $Gq_2 = 2q_2$. The loss reacts four times more strongly along q_1 than along q_2.

The Spectral Theorem Gives A Complete Coordinate System

For a real symmetric matrix $G \in \mathbb{R}^{d \times d}$, there exist orthonormal eigenvectors $q_1, \ldots, q_d$:

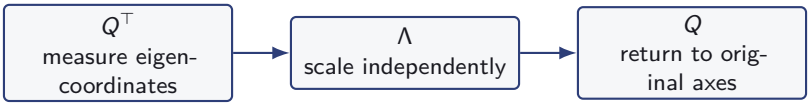
$$Q = [q_1 \ \cdots \ q_d], \qquad Q^\top Q = QQ^\top = I.$$

Collect the eigenvalues in

$$\Lambda = \mathrm{diag}(\lambda_1, \ldots, \lambda_d).$$

Then

$$\boxed{G = Q\Lambda Q^\top.}$$



Change Of Basis: The Vector Stays, Its Coordinates Change

Let $Q = [q_1 \ \cdots \ q_d]$ be orthogonal. For a coefficient vector θ, define its eigenbasis coordinates

$$\boxed{a = Q^\top \theta}.$$

Component j is

$$\boxed{a_j = q_j^\top \theta},$$

the signed coordinate of the same vector along axis q_j. Because $QQ^\top = I$,

$$\boxed{\theta = Qa = a_1 q_1 + \cdots + a_d q_d.}$$

Summary

Changing basis does not physically rotate the coefficient vector. It changes the axes used to describe that vector.

Rotate The Normal Equation Without Shorthand

Start from

$$X^\top X\widehat{\theta} = X^\top y,$$

substitute $X^\top X = Q\Lambda Q^\top$, and left-multiply both sides by $Q^\top$. Do not introduce a or b yet.

$$\begin{aligned} X^\top X\widehat{\theta} &= X^\top y && \\ Q\Lambda Q^\top\widehat{\theta} &= X^\top y && \text{spectral decomposition} \\ Q^\top Q\Lambda Q^\top\widehat{\theta} &= Q^\top X^\top y && \text{left-multiply by } Q^\top \\ \Lambda(Q^\top\widehat{\theta}) &= Q^\top X^\top y && \text{use } Q^\top Q = I. \end{aligned}$$

Both sides are now written in eigenvector coordinates.

Now Define a And b: Nothing Appears From Nowhere

The previous slide produced

$$\Lambda \underbrace{(Q^\top \widehat{\theta})}_{a} = \underbrace{Q^\top X^\top y}_{b}.$$

Thus

$$\boxed{a := Q^\top \widehat{\theta}}, \qquad \boxed{b := Q^\top X^\top y}, \qquad \boxed{\Lambda a = b}.$$

Coefficient coordinate

$$a_j = q_j^\top \widehat{\theta}.$$

Response information in direction q_j

$$\begin{aligned} b_j &= q_j^\top X^\top y \\ &= (Xq_j)^\top y. \end{aligned}$$

b_j measures the alignment between the observed response and the prediction pattern generated by moving coefficients along q_j.

A Coupled Matrix Equation Becomes Independent Scalar Equations

Because Λ is diagonal,

$$\Lambda a = b$$

means

$$\begin{bmatrix} \lambda_1 & & 0 \\ & \ddots & \\ 0 & & \lambda_d \end{bmatrix} \begin{bmatrix} a_1 \\ \vdots \\ a_d \end{bmatrix} = \begin{bmatrix} b_1 \\ \vdots \\ b_d \end{bmatrix}.$$

Direction by direction,

$$\boxed{\lambda_j a_j = b_j}, \qquad \boxed{a_j = \frac{b_j}{\lambda_j}} \quad (\lambda_j > 0).$$

Summary

The change of basis is useful because it replaces one coupled d-dimensional system with d transparent one-dimensional equations.

Why A Small Eigenvalue Makes Coefficients Sensitive

Perturb the rotated right-hand side from b to $b + \delta b$. Then

$$\Lambda(a + \delta a) = b + \delta b.$$

Subtract $\Lambda a = b$:

$$\Lambda \delta a = \delta b.$$

Direction j obeys

$$\boxed{\delta a_j = \frac{\delta b_j}{\lambda_j}}.$$

Large λ_j: a fixed data perturbation produces a small coefficient change.

Small λ_j: division by a small number can amplify the coefficient change.

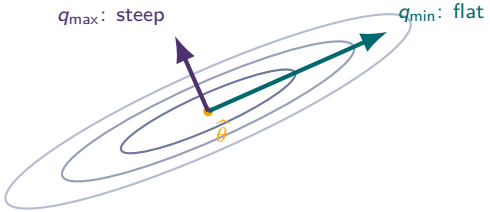
Summary

Predictions can remain similar because the same weak direction also satisfies $\|Xq_j\|_2^2 = \lambda_j$.

Eigenvalues Are The Curvatures Of The Loss Bowl

Near the least-squares optimum, move coefficients by Δ. In the eigenbasis,

$$\Delta = Qz, \qquad J(\widehat{\theta} + \Delta) - J(\widehat{\theta}) = \frac{1}{2} \sum_{j=1}^{d} \lambda_j z_j^2.$$



$$\boxed{\kappa(X^\top X) = \lambda_{\max}/\lambda_{\min}}.$$

A large condition number means very unequal curvature and slow first-order optimization along the flat direction.

Bridge To PCA: Same Gram Pattern, Different Question

Regression

$$H = X^\top X.$$

Acts on a coefficient move q. Eigenvalue measures loss curvature and coefficient sensitivity. A response y is present.

PCA

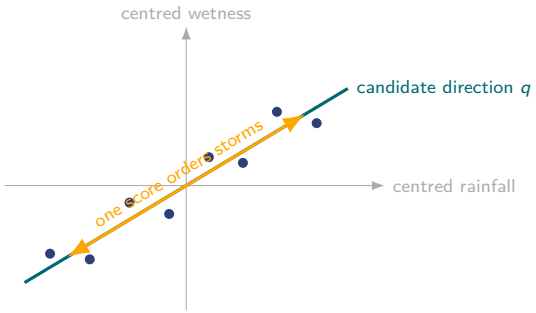
$$C = \frac{1}{n-1} X_c^\top X_c.$$

Acts on a feature direction q. Eigenvalue will measure projected data variance. No response y is used.

Regression asks how the loss reacts. PCA asks how the data vary.

PCA Motivation: Can Two Features Become One Useful Score?

Each point is one storm described by centred rainfall and centred wetness.



Compression question

Can we replace

$$x_i = \begin{bmatrix} \text{rainfall}_i \\ \text{wetness}_i \end{bmatrix}$$

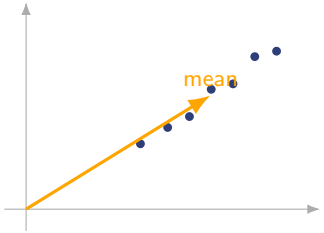
with one scalar z_i while retaining the main differences among storms?

Why Centre The Data Before PCA?

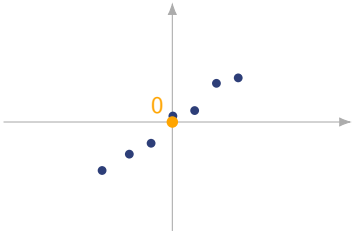
For each observation, subtract the feature mean:

$$\boxed{x_{i,c} = x_i - \bar{x}}, \qquad \frac{1}{n}\sum_{i=1}^{n} x_{i,c} = 0.$$

Before centring



After centring



Without centring, distance from the origin is confused with variation around the mean.

Step 1: Choose A Direction And Fix Its Length

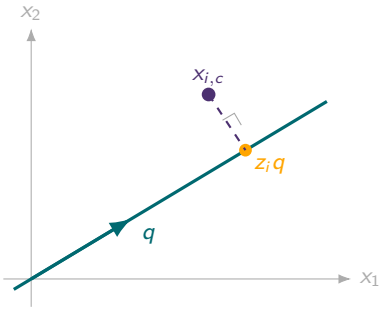
Choose a candidate direction $q \in \mathbb{R}^d$. The line is $\text{span}(q)$, so q and cq describe the same line.
To prevent scale from changing the answer, require

$$\boxed{q^\top q = 1.}$$

For centred observation $x_{i,c}$, write its projection as $\widehat{x}_{i,c} = z_i q$. Perpendicularity gives

$$q^\top (x_{i,c} - z_i q) = 0$$
$$q^\top x_{i,c} - z_i \underbrace{q^\top q}_{1} = 0$$

$$\Longrightarrow \quad \boxed{z_i = x_{i,c}^\top q.}$$



Summary

q is a direction; z_i is observation i's signed coordinate on that direction. They are not the same object.

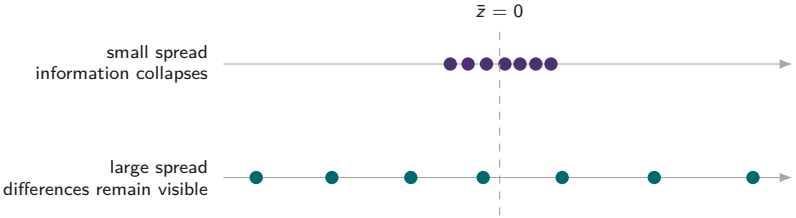
Step 2: Project Every Observation And Measure The Spread

Stack the coordinates $z_i = x_{i,c}^\top q$ for all n observations:

$$\boxed{z = X_c q}, \qquad (n \times d)(d \times 1) = n \times 1.$$

Because each column of X_c is centred,

$$\mathbf{1}^\top X_c = 0^\top \quad \Longrightarrow \quad \bar{z} = \frac{1}{n}\mathbf{1}^\top z = \frac{1}{n}\mathbf{1}^\top X_c q = 0.$$



For a fixed unit direction q, PCA compares the spread of the projected coordinates $z_1, \ldots, z_n$.

Step 3: The Objective And Constraint Now Fit Together

Start from the variance of the projected coordinates. Since $\bar{z} = 0$,

$$\begin{aligned}\mathrm{Var}(z) &= \frac{1}{n-1}\sum_{i=1}^{n}(z_i - \bar{z})^2 = \frac{1}{n-1}\sum_{i=1}^{n} z_i^2 \\ &= \frac{1}{n-1} z^\top z = \frac{1}{n-1}(X_c q)^\top (X_c q) \\ &= q^\top \underbrace{\left(\frac{1}{n-1} X_c^\top X_c\right)}_{C} q.\end{aligned}$$

Define the feature covariance matrix

$$\boxed{C = \frac{1}{n-1} X_c^\top X_c}.$$

The first PCA direction is therefore

$$\boxed{q_1 = \arg\max_{q \in \mathbb{R}^d} q^\top C q \quad \text{subject to} \quad q^\top q = 1.}$$

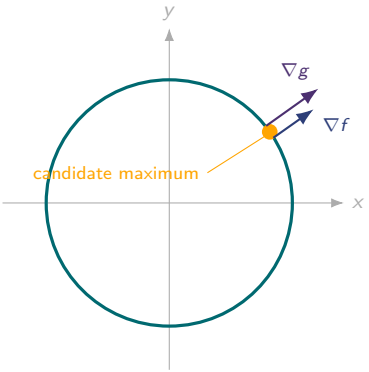
Objective $q^\top C q$**:** retain the largest projected variance.

Constraint $q^\top q = 1$**:** prevent scale from manufacturing variance.

Lagrange Multipliers Enforce The Constraint

Before applying the method to PCA, consider

$$\max_{x,y} \; f(x,y) = x^2 y \quad \text{subject to} \quad x^2 + y^2 = 1.$$



Write the constraint as

$$g(x,y) = x^2 + y^2 - 1 = 0.$$

At a smooth constrained optimum, the objective gradient is parallel to the constraint normal:

$$\boxed{\nabla f = \lambda \nabla g}.$$

Equivalently,

$$\boxed{\mathcal{L}(x,y,\lambda) = f(x,y) - \lambda g(x,y)}$$

and all partial derivatives are set to zero.

Solve $\max x^2 y$ On The Unit Circle Step By Step

For

$$\mathcal{L} = x^2 y - \lambda(x^2 + y^2 - 1),$$

derive the stationary equations, separate the $x = 0$ and $x \neq 0$ cases, and identify the global maximum.

$$\frac{\partial \mathcal{L}}{\partial x} = 2xy - 2\lambda x = 0,$$
$$\frac{\partial \mathcal{L}}{\partial y} = x^2 - 2\lambda y = 0,$$
$$\frac{\partial \mathcal{L}}{\partial \lambda} = -(x^2 + y^2 - 1) = 0.$$

If $x = 0$, then $y = \pm 1$ and $f = 0$.
If $x \neq 0$, the first equation gives $\lambda = y$. The second gives $x^2 = 2y^2$. With $x^2 + y^2 = 1$,

$$3y^2 = 1, \quad y = \pm\frac{1}{\sqrt{3}}, \quad x = \pm\sqrt{\frac{2}{3}}.$$

The global maximum is

$$\boxed{f_{\max} = \frac{2}{3\sqrt{3}}}$$

at $y = 1/\sqrt{3}$ and $x = \pm\sqrt{2/3}$.

The PCA Constraint Produces An Eigenvalue Equation

Apply the same method to

$$\max_{q} \; q^\top C q \qquad \text{subject to} \qquad q^\top q = 1.$$

The Lagrangian is

$$\mathcal{L}(q, \lambda) = q^\top C q - \lambda(q^\top q - 1).$$

Because C is symmetric,

$$\nabla_q \mathcal{L} = 2Cq - 2\lambda q = 0$$

$$\Longrightarrow \boxed{Cq = \lambda q.}$$

At a unit eigenvector q_j,

$$q_j^\top C q_j = q_j^\top (\lambda_j q_j) = \lambda_j.$$

Thus maximizing projected variance means selecting

$$\boxed{q^* = q_1, \qquad \lambda_1 = \max_j \lambda_j.}$$

The candidate was named q; after optimization, the winning direction is the eigenvector q_1.

Keeping All r Informative Directions Is Exact

Let $r = \operatorname{rank}(X_c)$ and collect all eigenvectors associated with positive eigenvalues:

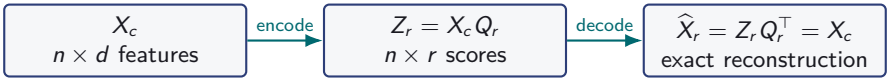
$$Q_r = [q_1 \ \cdots \ q_r] \in \mathbb{R}^{d \times r}.$$

The score matrix is

$$\boxed{Z_r = X_c Q_r \in \mathbb{R}^{n \times r}.}$$

Reconstruct by returning from score coordinates to the original feature axes:

$$\boxed{\widehat{X}_r = Z_r Q_r^\top = X_c Q_r Q_r^\top = X_c.}$$



If $r = d$, this is only a lossless change of coordinates. If $r < d$, the data already contain exact redundancy, so reducing from d to r is lossless.

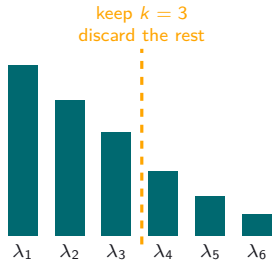
Keeping Only $k < r$ Directions Compresses With Loss

Choose the first k eigenvectors, $Q_k = [q_1 \ \cdots \ q_k]$:

$$\underbrace{Z_k = X_c Q_k}_{n \times k \text{ compressed scores}} \quad , \quad \underbrace{\widehat{X}_k = Z_k Q_k^\top}_{n \times d \text{ reconstruction}} .$$

The discarded eigenvalues give the sample-scaled squared reconstruction loss:

$$\boxed{\frac{1}{n-1}\|X_c - \widehat{X}_k\|_F^2 = \sum_{j=k+1}^{r} \lambda_j > 0.}$$



For genuine compression $k < r$, some reconstruction information is necessarily lost.

PCA makes this squared loss as small as possible among rank-k linear representations. Large discarded eigenvalues mean substantial loss; tiny ones mean mild loss.

What PCA Preserves, What It May Lose, And Why ML Cares

What PCA guarantees

For a chosen k, PCA retains the largest possible total variance and minimizes squared linear reconstruction error.

$$\text{retained fraction} = \frac{\sum_{j=1}^{k} \lambda_j}{\sum_{j=1}^{r} \lambda_j}.$$

What PCA does not guarantee

Low-variance directions may still contain labels, rare events, or scientifically important signals. Maximum variance is not the same as maximum predictive value.

The same eigen-algebra answers two ML questions

Linear regression: eigenpairs of $X^\top X$ reveal coefficient sensitivity, identifiability, and loss curvature.

PCA: eigenpairs of $C = \frac{1}{n-1} X_c^\top X_c$ reveal feature-space variance directions and the cost of compression.

Summary

Eigenvectors identify special directions. Eigenvalues quantify how strongly the data express those directions.

Next: logistic regression, where probability replaces the continuous response but optimization remains central.